

Meeting 1022

黃丞暘

國立東華大學應數統計所

October 22, 2025

目錄

1 研究動機

2 文獻回顧 I

3 研究方法

4 研究資料

5 研究結果

6 參考資料

研究動機

Semiconductor Manufacturing Process

- ▶ Robust performance can be trust
- ▶ High dimension
- ▶ Find the $y = \hat{f}(x)$

研究動機

Semiconductor Manufacturing Process

- ▶ Robust performance can be trust
- ▶ High dimension
- ▶ Find the $y = \hat{f}(x)$

研究動機

Semiconductor Manufacturing Process

- ▶ Robust performance can be trust
- ▶ High dimension
- ▶ Find the $y = \hat{f}(x)$

文獻回顧 I

► Lai (2002)

■ 方法

- 1 Data preprocessing
- 2 KNN-imputation
- 3 Random forest features selection
- 4 imbalance - SMOTE to 1:1

文獻回顧 I

► Lai (2002)

■ 方法

- 1 Data preprocessing
- 2 KNN-imputation
- 3 Random forest features selection
- 4 imbalance - SMOTE to 1:1

文獻回顧 I

► Lai (2002)

■ 方法

- 1 Data preprocessing
- 2 KNN-imputation
- 3 Random forest features selection
- 4 imbalance - SMOTE to 1:1

文獻回顧 I

► Lai (2002)

■ 方法

- 1 Data preprocessing
- 2 KNN-imputation
- 3 Random forest features selection
- 4 imbalance - SMOTE to 1:1

文獻回顧 I

► Lai (2002)

■ 方法

- 1 Data preprocessing
- 2 KNN-imputation
- 3 Random forest features selection
- 4 imbalance - SMOTE to 1:1

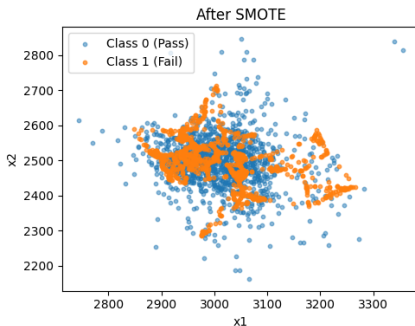
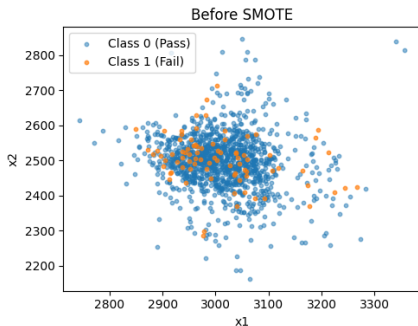
文獻回顧 I

► Lai (2002)

■ 方法

- 1 Data preprocessing
- 2 KNN-imputation
- 3 Random forest features selection
- 4 imbalance - SMOTE to 1:1

文獻回顧 I-方法 (SMOTE)



文獻回顧 I

► Lai (2002)

■ 方法

- 1 Data preprocessing
- 2 KNN-imputation
- 3 Random forest features selection
- 4 imbalance - SMOTE to 1:1
- 5 0.8 training, 0.2 testing by decision tree

■ 結果

- 1 Best: ACC 86.94%, FAR 9.22%
- 2 反查 Fail 節點

文獻回顧 I

► Lai (2002)

■ 方法

- 1 Data preprocessing
- 2 KNN-imputation
- 3 Random forest features selection
- 4 imbalance - SMOTE to 1:1
- 5 0.8 training, 0.2 testing by decision tree

■ 結果

- 1 Best: ACC 86.94%, FAR 9.22%
- 2 反查 Fail 節點

文獻回顧 I

► Lai (2002)

■ 方法

- 1 Data preprocessing
- 2 KNN-imputation
- 3 Random forest features selection
- 4 imbalance - SMOTE to 1:1
- 5 0.8 training, 0.2 testing by decision tree

■ 結果

- 1 Best: ACC 86.94%, FAR 9.22%
- 2 反查 Fail 節點

文獻回顧 I

► Lai (2002)

■ 方法

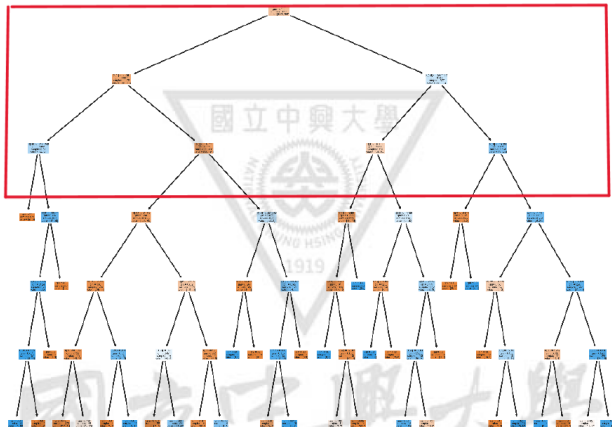
- 1 Data preprocessing
- 2 KNN-imputation
- 3 Random forest features selection
- 4 imbalance - SMOTE to 1:1
- 5 0.8 training, 0.2 testing by decision tree

■ 結果

- 1 Best: ACC 86.94%, FAR 9.22%
- 2 反查 Fail 節點

文獻回顧 I

反查節點



文獻回顧 I

反查節點

表 4.3: 透過決策樹找出異常原因範例

Fail Case	High Concerned Factors
Data group(797)	$X(54) \leq 4.57$, $X(121) > 2.24$, $X(59) \leq 28.18$, $X(118) \leq 0.29$, $X(95) \leq -0.00$, $X(282) \leq 0.05$
Data group(929)	$X(54) \leq 4.57$, $X(121) > 2.24$, $X(59) \leq 28.18$, $X(118) \leq 0.29$, $X(95) \leq -0.00$, $X(282) \leq 0.05$
Data group(926)	$X(54) > 4.57$, $X(19) > -5464.47$, $X(110) > 15.73$, $X(153) > 0.09$, $X(31) > 8.53$, $X(236) > 0.01$
Data group(154)	$X(54) > 4.57$, $X(19) \leq -5464.47$, $X(59) > 16.34$, $X(69) \leq -0.02$, $X(244) > 0.52$
Data group(277)	$X(54) > 4.57$, $X(19) > -5464.47$, $X(110) > 15.73$, $X(153) > 0.09$, $X(31) > 8.53$, $X(236) > 0.01$

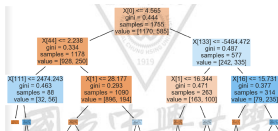


圖 4.12: 最佳化決策樹前三節點

研究方法

► 前處理

■ vote in features selection

► 機器學習模型

■ XGboost, RandomForest, DecisionTree, SVM



研究方法

► 前處理

■ vote in features selection

► 機器學習模型

■ XGboost, RandomForest, DecisionTree, SVM



研究方法

▶ 前處理

- vote in features selection

▶ 機器學習模型

- XGboost, RandomForest, DecisionTree, SVM



研究方法

► 前處理

- vote in features selection

► 機器學習模型

- XGboost, RandomForest, DecisionTree, SVM



研究方法

▶ 前處理

- vote in features selection

▶ 機器學習模型

- XGboost, RandomForest, DecisionTree, SVM



研究流程

資料前處理 → RF 取 order → Train/Test 分割 → 5-fold → 4 模型
×top-k → 折內選 k → 跨折選全域 k → 重訓 → 測試評估

研究資料

SECOM

▶ 樣本數 1567

▶ 變數 Time / x1 / x2 / ... / x590 / PassFail

■ Features: 591, Y: PassFail (binary classification)

■ Time: 2008/01/08 - 2008/12/10

■ PassFail: Pass $\rightarrow$ 0, Fail $\rightarrow$ 1 (14:1)

■ Missing values: 41951, every cols at least 1

研究資料

SECOM

- ▶ 樣本數 1567
- ▶ 變數 Time / x1 / x2 / ... / x590 / PassFail
 - Features: 591, Y: PassFail (binary classification)
 - Time: 2008/01/08 - 2008/12/10
 - PassFail: Pass $\rightarrow$ 0, Fail $\rightarrow$ 1 (14:1)
 - Missing values: 41951, every cols at least 1

研究資料

SECOM

- ▶ 樣本數 1567
- ▶ 變數 Time / x1 / x2 / ... / x590 / PassFail
 - Features: 591, Y: PassFail (binary classification)
 - Time: 2008/01/08 - 2008/12/10
 - PassFail: Pass $\rightarrow$ 0, Fail $\rightarrow$ 1 (14:1)
 - Missing values: 41951, every cols at least 1

研究資料

SECOM

- ▶ 樣本數 1567
- ▶ 變數 Time / x1 / x2 / ... / x590 / PassFail
 - Features: 591, Y: PassFail (binary classification)
 - Time: 2008/01/08 - 2008/12/10
 - PassFail: Pass $\rightarrow$ 0, Fail $\rightarrow$ 1 (14:1)
 - Missing values: 41951, every cols at least 1

研究資料

SECOM

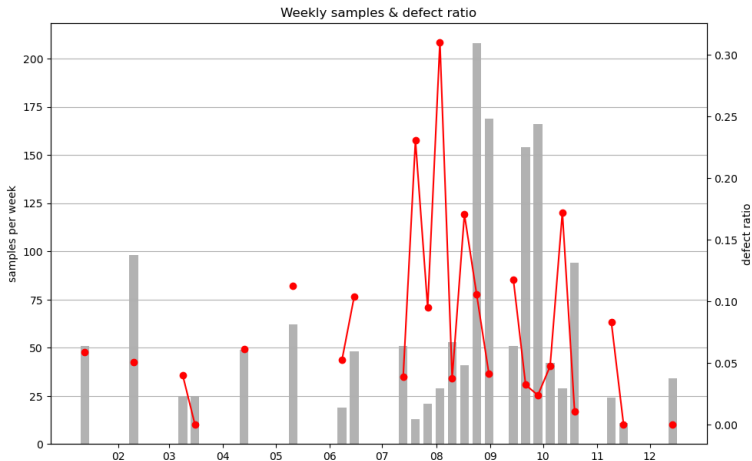
- ▶ 樣本數 1567
- ▶ 變數 Time / x1 / x2 / ... / x590 / PassFail
 - Features: 591, Y: PassFail (binary classification)
 - Time: 2008/01/08 - 2008/12/10
 - PassFail: Pass $\rightarrow$ 0, Fail $\rightarrow$ 1 (14:1)
 - Missing values: 41951, every cols at least 1

研究資料

SECOM

- ▶ 樣本數 1567
- ▶ 變數 Time / x1 / x2 / ... / x590 / PassFail
 - Features: 591, Y: PassFail (binary classification)
 - Time: 2008/01/08 - 2008/12/10
 - PassFail: Pass $\rightarrow$ 0, Fail $\rightarrow$ 1 (14:1)
 - Missing values: 41951, every cols at least 1

研究資料



研究結果 I

$f(X) = f(\text{significant linear features}, (X_1)^2, (X_2)^2, \dots, (X_p)^2)$

中位數插補

隨機森林

ACC=0.9076, FAR=0.0375

	Pred 0	Pred 1
True 0	282	11
True 1	18	3

XGboost

ACC=0.9236 FAR=0.0102

	Pred 0	Pred 1
True 0	290	3
True 1	21	0

KNN 插補

隨機森林

ACC=0.8981 FAR=0.0478

	Pred 0	Pred 1
True 0	279	14
True 1	18	3

XGboost

ACC=0.9268 FAR=0.0068

	Pred 0	Pred 1
True 0	291	2
True 1	21	0

研究結果 II

最終 Testing Set 結果 (各模型用 5 折挑出的最佳 k)

model	k	mean_val_acc	tr_acc	te_acc	te_FAR
DT	75	0.785297	0.874359	0.678344	0.303754
RF	185	0.901833	0.969231	0.901274	0.054608
SVC	195	0.916204	0.993590	0.910828	0.030717
XGB	180	0.928972	1.000000	0.920382	0.017065

研究結果 II

DT

	Pred 0	Pred 1
True 0	204	89
True 1	12	9

RF

	Pred 0	Pred 1
True 0	277	16
True 1	15	6

SVC

	Pred 0	Pred 1
True 0	284	9
True 1	19	2

XGB

	Pred 0	Pred 1
True 0	288	5
True 1	20	1

參考資料



Li-Fu Lai (2022)

使用 KNN 和決策樹方法對填補缺失值的製造數據進行失效預測和原因追蹤分析

The End

Questions & Comments